SubQueries

Question 1: Retrieve the names of employees who earn more than the average salary of all employees.

SQL Query:

```
USE subqueries_db;


SELECT name, salary

FROM Employee

WHERE salary > (

   SELECT AVG(salary)

   FROM Employee

);
```

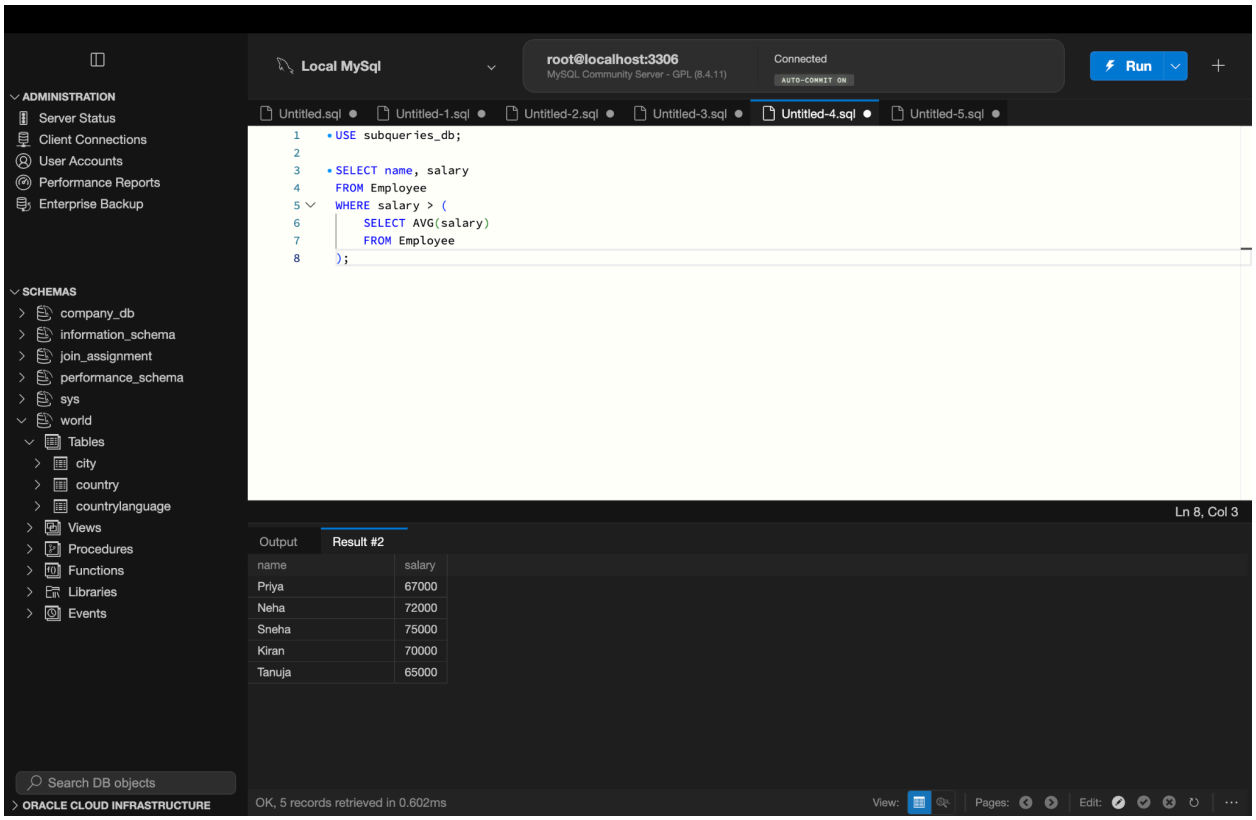
Explanation:

The subquery calculates the average salary of all employees. The outer query retrieves the names and salaries of employees whose salaries are greater than the average.

Result:

The average salary of all employees is ₹64,800. Five employees earn more than the average salary.

Practical proof :



Question 2: Find the employees who belong to the department with the highest average salary.

SQL Query:

```
SELECT name, department_id, salary
FROM Employee
WHERE department_id IN (
    SELECT department_id
    FROM Employee
    GROUP BY department_id
    HAVING AVG(salary) = (
        SELECT MAX(avg_salary)
        FROM (
            SELECT AVG(salary) AS avg_salary
            FROM Employee
            GROUP BY department_id
        ) AS dept_avg
```

```
    )
);
```

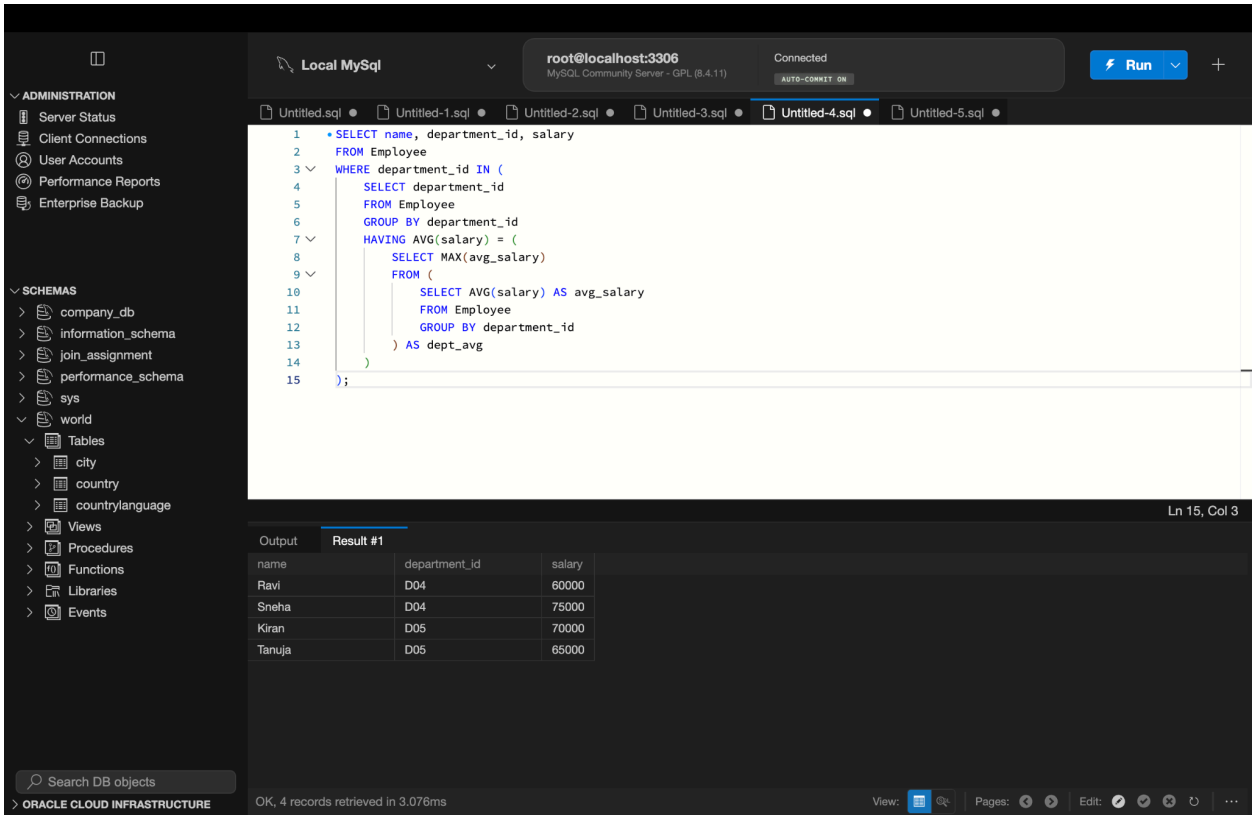
Explanation:

The innermost subquery calculates the average salary of each department. The next subquery identifies the highest departmental average salary. The outer query retrieves employees who belong to departments with that average salary.

Result:

Departments D04 (HR) and D05 (IT) have the highest average salary of ₹67,500. Therefore, four employees belong to these departments.

Practical proof :



Question 3: List all employees who have made at least one sale.

SQL Query:

```
USE subqueries_db;

SELECT name
FROM Employee
WHERE emp_id IN (
    SELECT emp_id
    FROM Sales
);
```

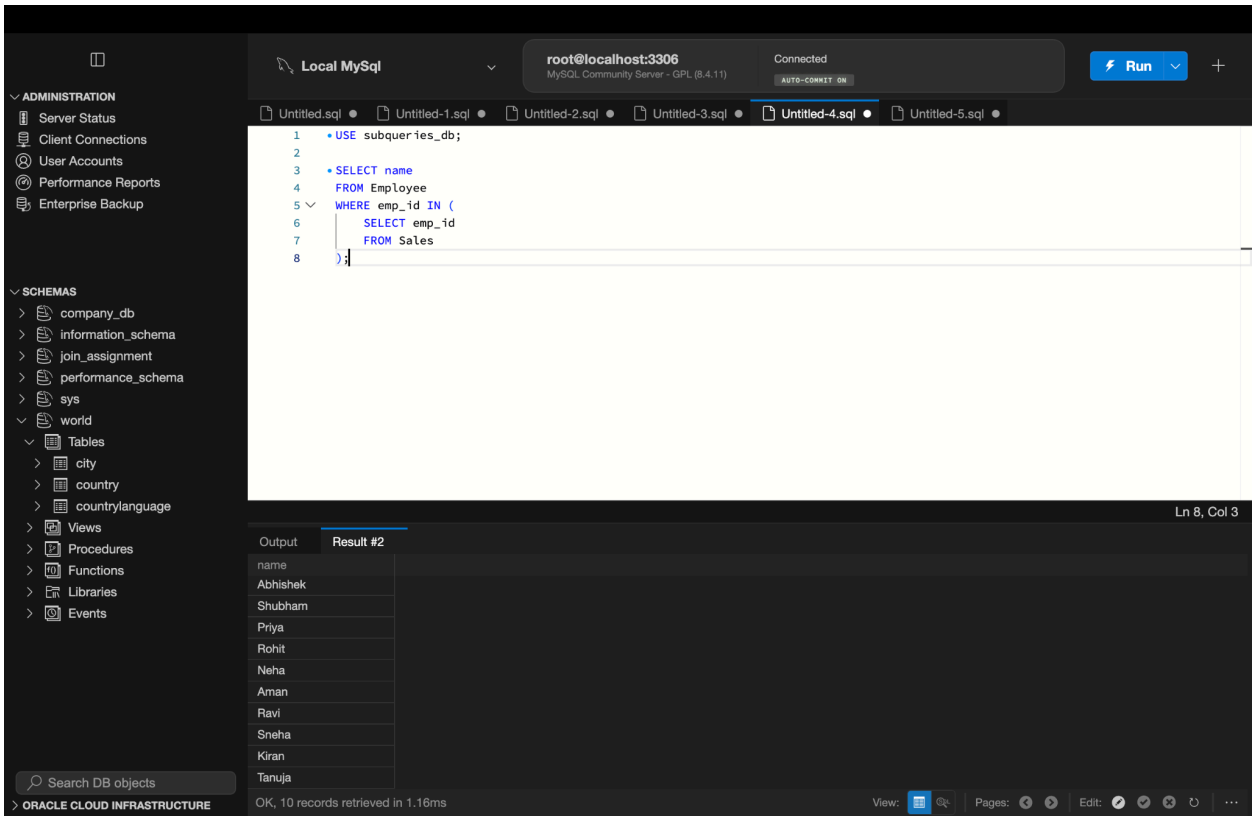
Explanation:

The subquery retrieves the employee IDs from the Sales table. The outer query uses the `IN` operator to retrieve the names of employees whose IDs appear in the Sales table.

Result:

All 10 employees have made at least one sale.

Practical proof :



Question 4: Find the employee with the highest sale amount.

SQL Query:

```
USE subqueries_db;

SELECT name, sale_amount
FROM Employee
JOIN Sales ON Employee.emp_id = Sales.emp_id
WHERE sale_amount = (
    SELECT MAX(sale_amount)
    FROM Sales
);
```

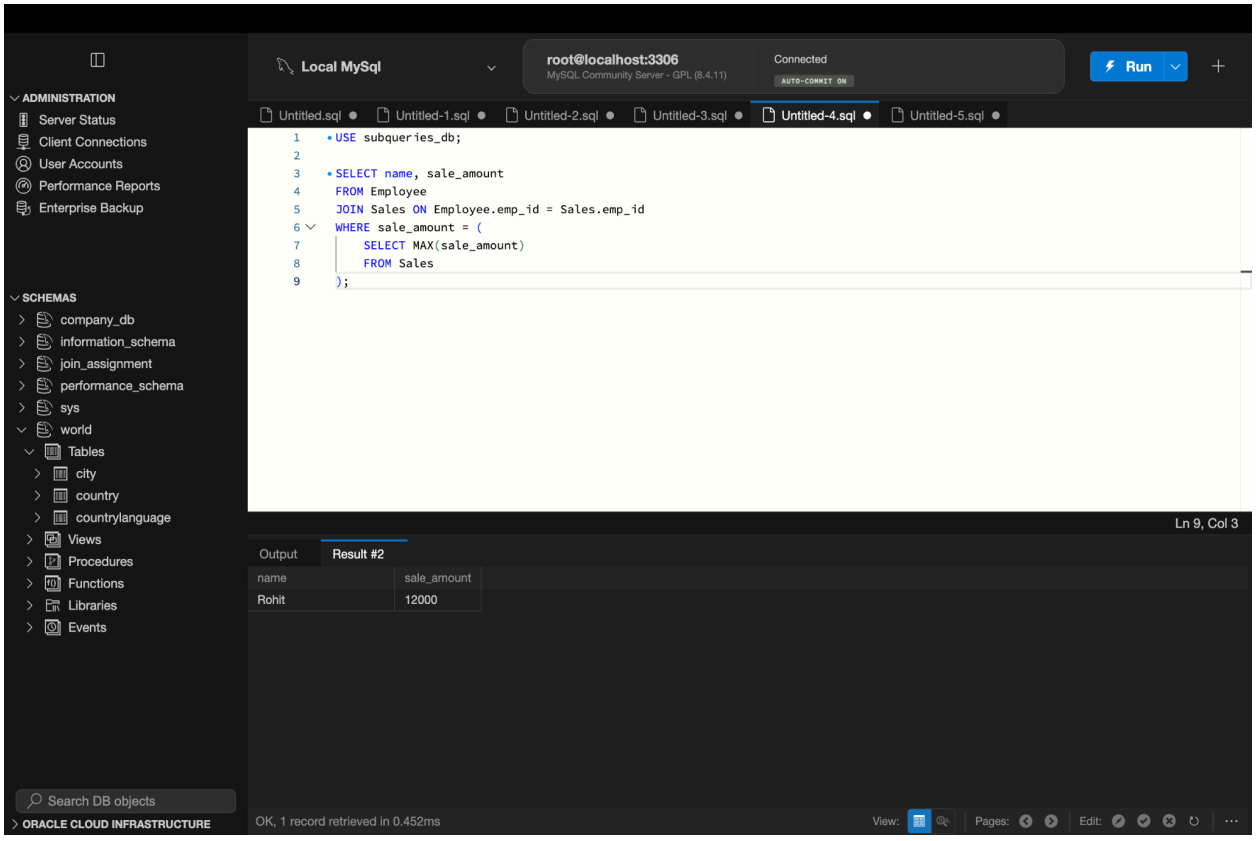
Explanation:

The innermost subquery finds the highest sale amount recorded in the Sales table. The next subquery identifies the employee ID associated with that sale. The outer query retrieves the employee's name and sale amount.

Result:

Rohit made the highest sale, amounting to ₹12,000.

Practical proof :



Question 5: Retrieve the names of employees whose salaries are higher than Shubham’s salary.

SQL Query:

```
USE subqueries_db;

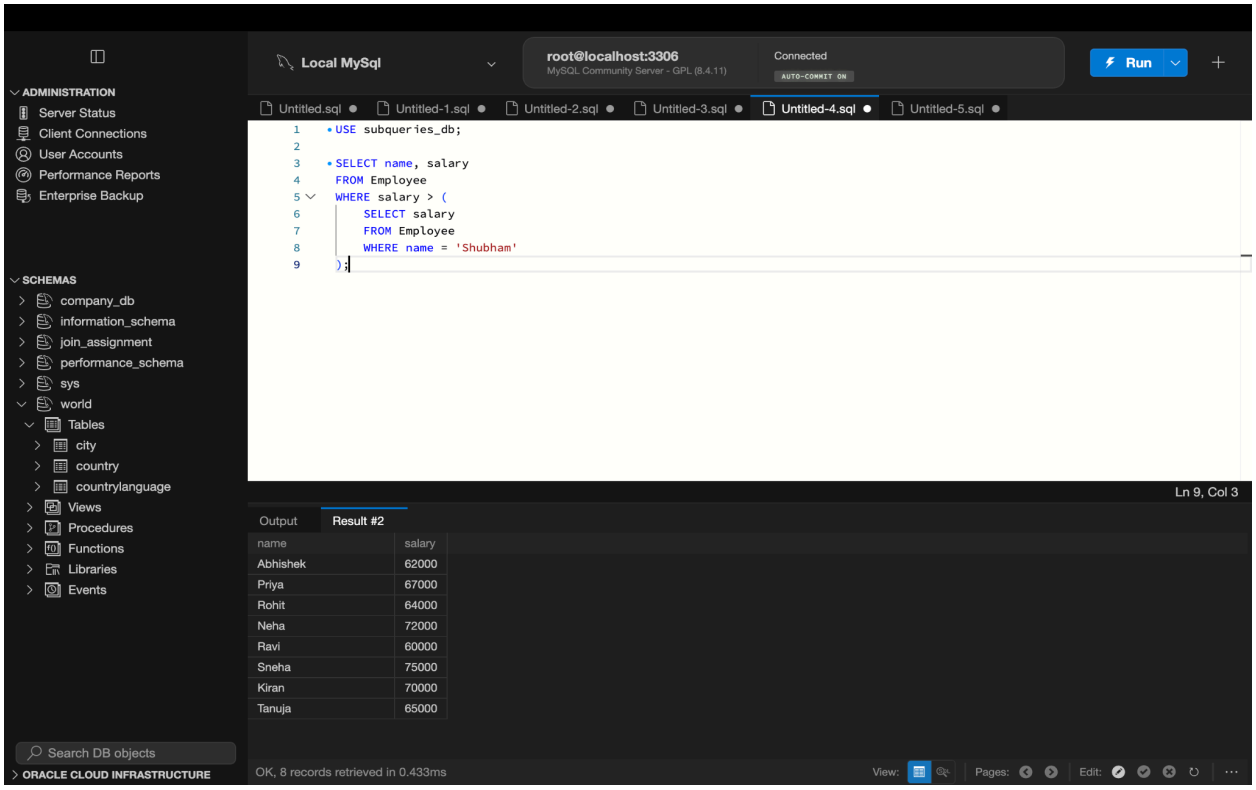
SELECT name, salary
FROM Employee
WHERE salary > (
    SELECT salary
    FROM Employee
    WHERE name = 'Shubham'
);
```

Explanation:

The subquery retrieves Shubham’s salary, which is ₹58,000. The main query displays the names and salaries of employees earning more than ₹58,000.

Result: Eight employees earn more than Shubham.

Practical Proof:



Question 6: Find employees who work in the same department as Abhishek.

SQL Query:

```
USE subqueries_db;

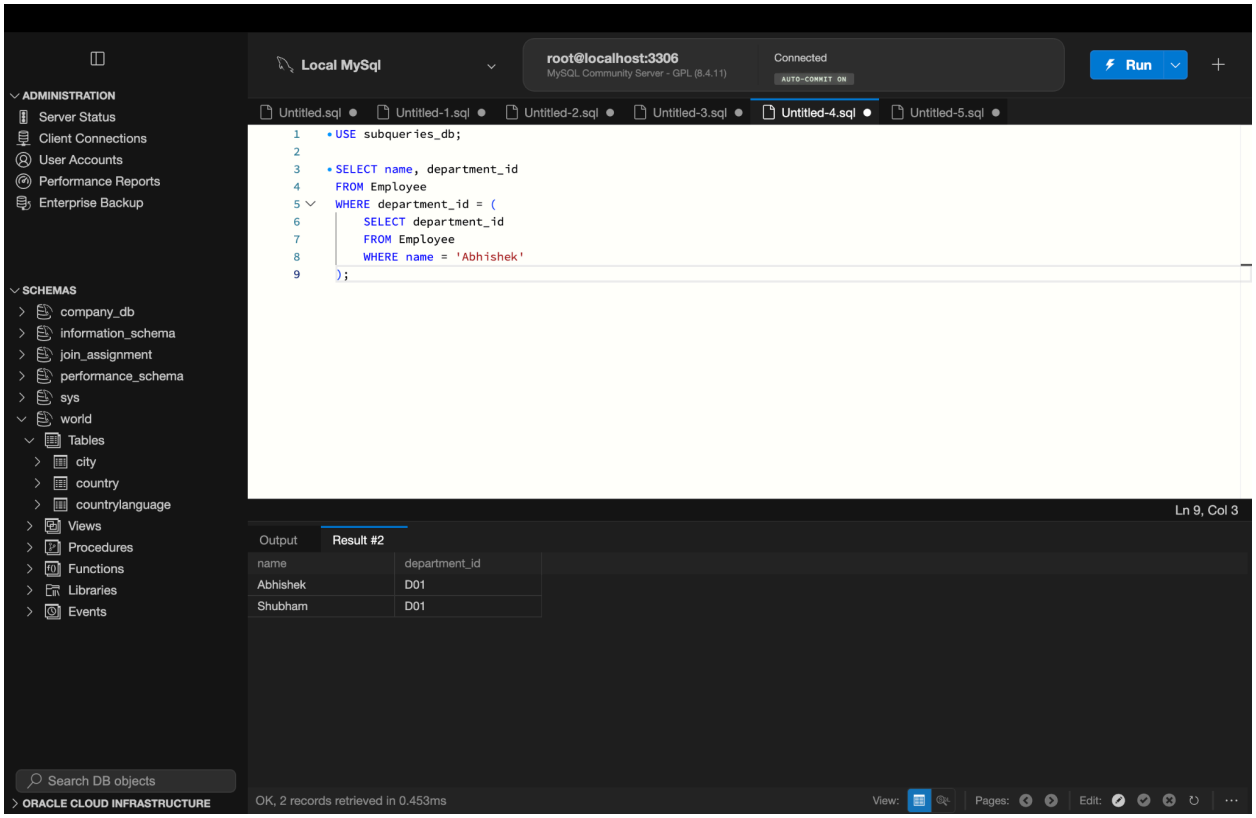

SELECT name, department_id
FROM Employee
WHERE department_id = (
   SELECT department_id
   FROM Employee
   WHERE name = 'Abhishek'
);
```

Explanation:

The subquery retrieves Abhishek's department ID, which is D01. The main query displays all employees working in the same department.

Result: Abhishek and Shubham work in department D01 (Sales).

Practical Proof:



Question 7: List departments with at least one employee earning more than ₹60,000.

SQL Query:

USE subqueries_db;

SELECT department_id, department_name

FROM Department

WHERE department_id IN (

SELECT department_id

FROM Employee

WHERE salary > 60000

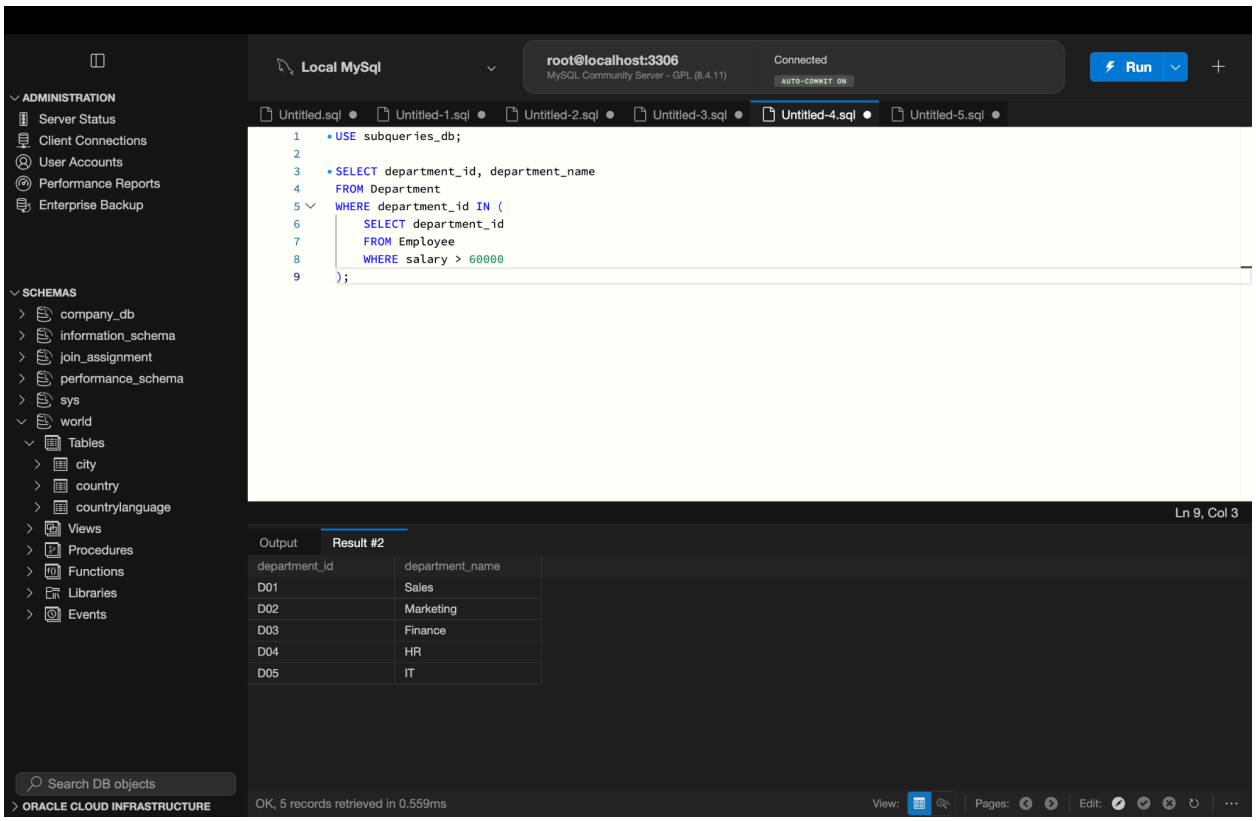
);

Explanation:

The subquery retrieves the department IDs of employees earning more than ₹60,000. The main query displays the corresponding department IDs and names.

Result: All five departments have at least one employee earning more than ₹60,000.

Practical Proof:



Question 8: Find the department name of the employee who made the highest sale.

SQL Query:

USE subqueries_db;

SELECT Department.department_name

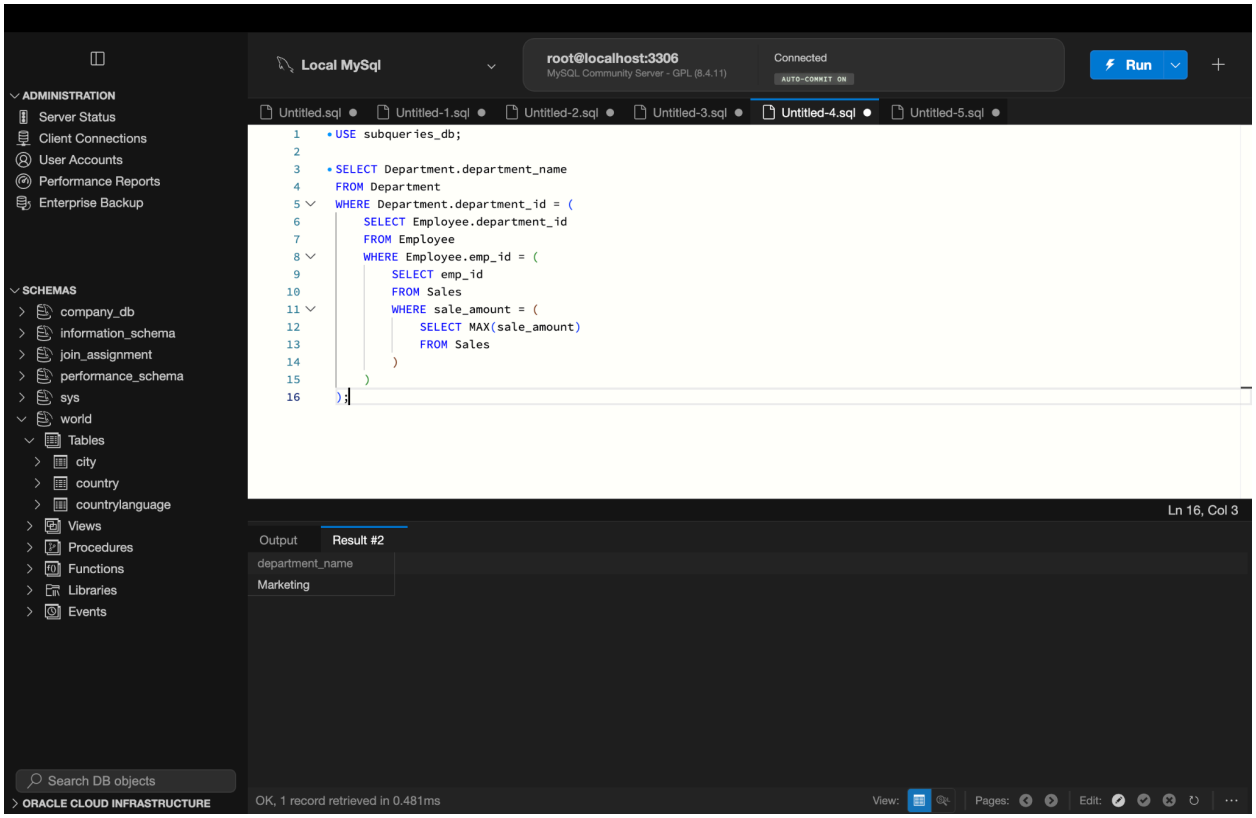
```
FROM Department
WHERE Department.department_id = (
   SELECT Employee.department_id
   FROM Employee
   WHERE Employee.emp_id = (
      SELECT emp_id
      FROM Sales
      WHERE sale_amount = (
         SELECT MAX(sale_amount)
         FROM Sales
      )
   )
);
```

Explanation:

The innermost subquery finds the highest sale amount, which is ₹12,000. The next subquery identifies the employee who made that sale, Rohit (employee ID 104). The main query retrieves his department name.

Result: Rohit belongs to the Marketing department, which recorded the highest individual sale of ₹12,000.

Practical Proof:



Question 9: Retrieve employees who made sales greater than the average sale amount.

SQL Query:

USE subqueries_db;

SELECT Employee.name, Sales.sale_amount

FROM Employee

JOIN Sales ON Employee.emp_id = Sales.emp_id

WHERE Sales.sale_amount > (

SELECT AVG(sale_amount)

FROM Sales

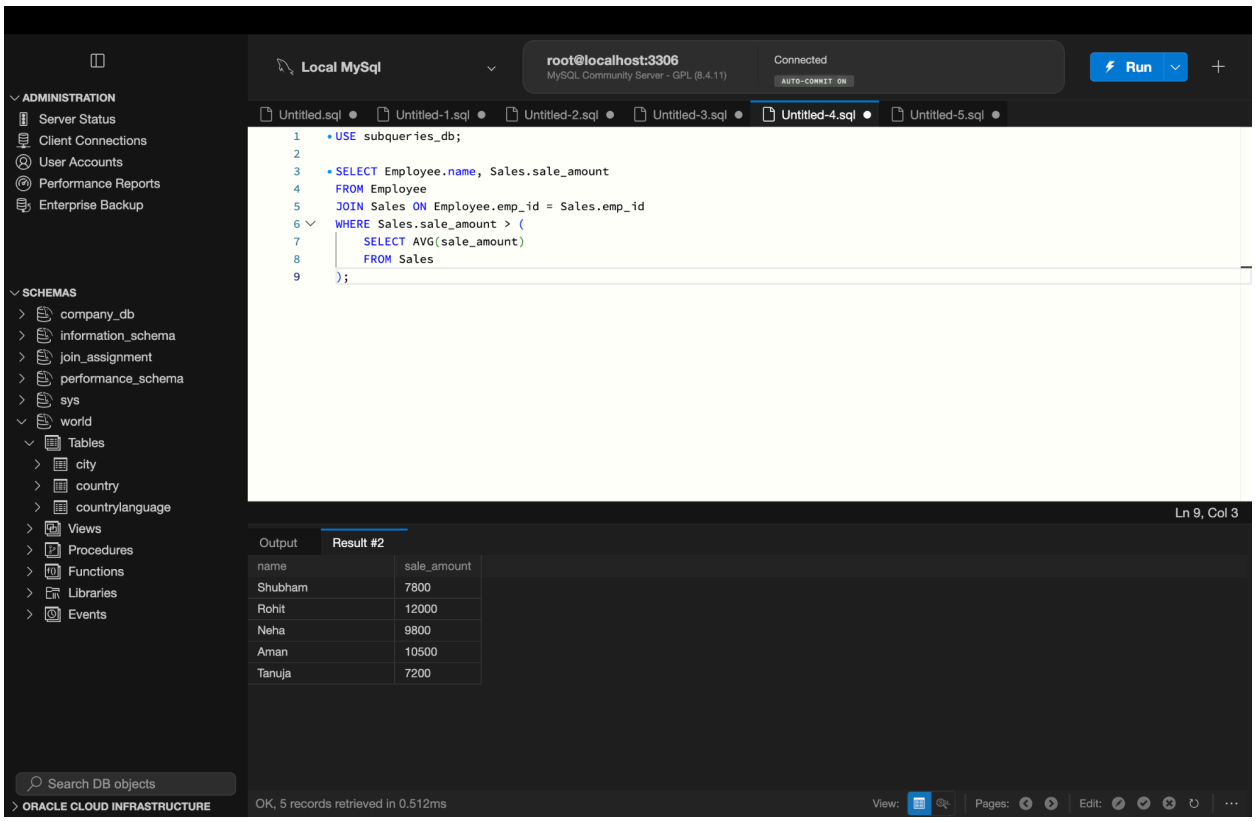
);

Explanation:

The subquery calculates the average sale amount, which is ₹7,070. The main query displays the names of employees whose sale amounts are greater than the average.

Result: Six employees made sales greater than the average sale amount.

Practical Proof :



Question 10: Find the total sales made by employees earning more than the average salary.

SQL Query:

USE subqueries_db;

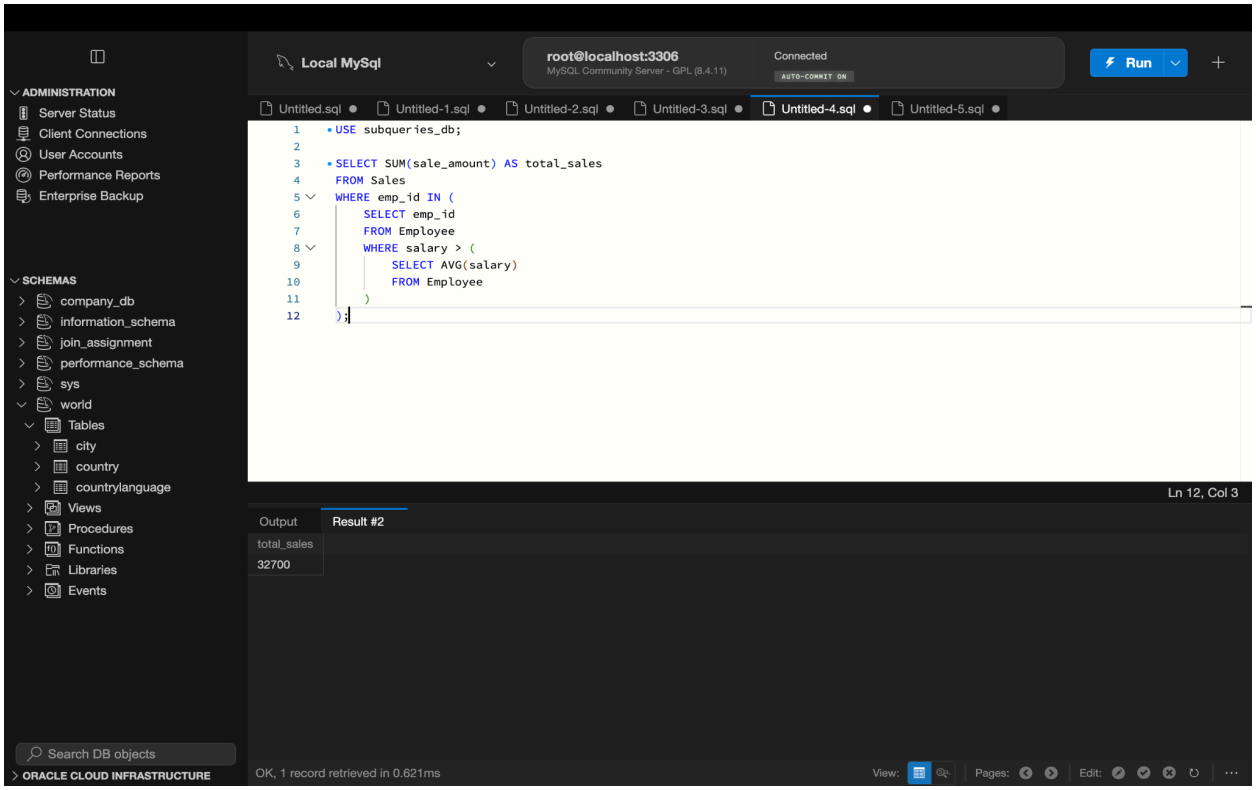
```
SELECT SUM(sale_amount) AS total_sales
FROM Sales
WHERE emp_id IN (
    SELECT emp_id
    FROM Employee
    WHERE salary > (
        SELECT AVG(salary)
        FROM Employee
    )
);
```

Explanation:

The innermost subquery calculates the average employee salary, which is ₹64,800. The next subquery identifies employees earning more than the average salary. The main query calculates the total sales made by those employees.

Result: The total sales made by employees earning more than the average salary is ₹27,000.

Practical Proof:



Question 11: Find employees who have not made any sales.

SQL Query:

```
USE subqueries_db;


SELECT name, emp_id

FROM Employee

WHERE emp_id NOT IN (

   SELECT emp_id

   FROM Sales

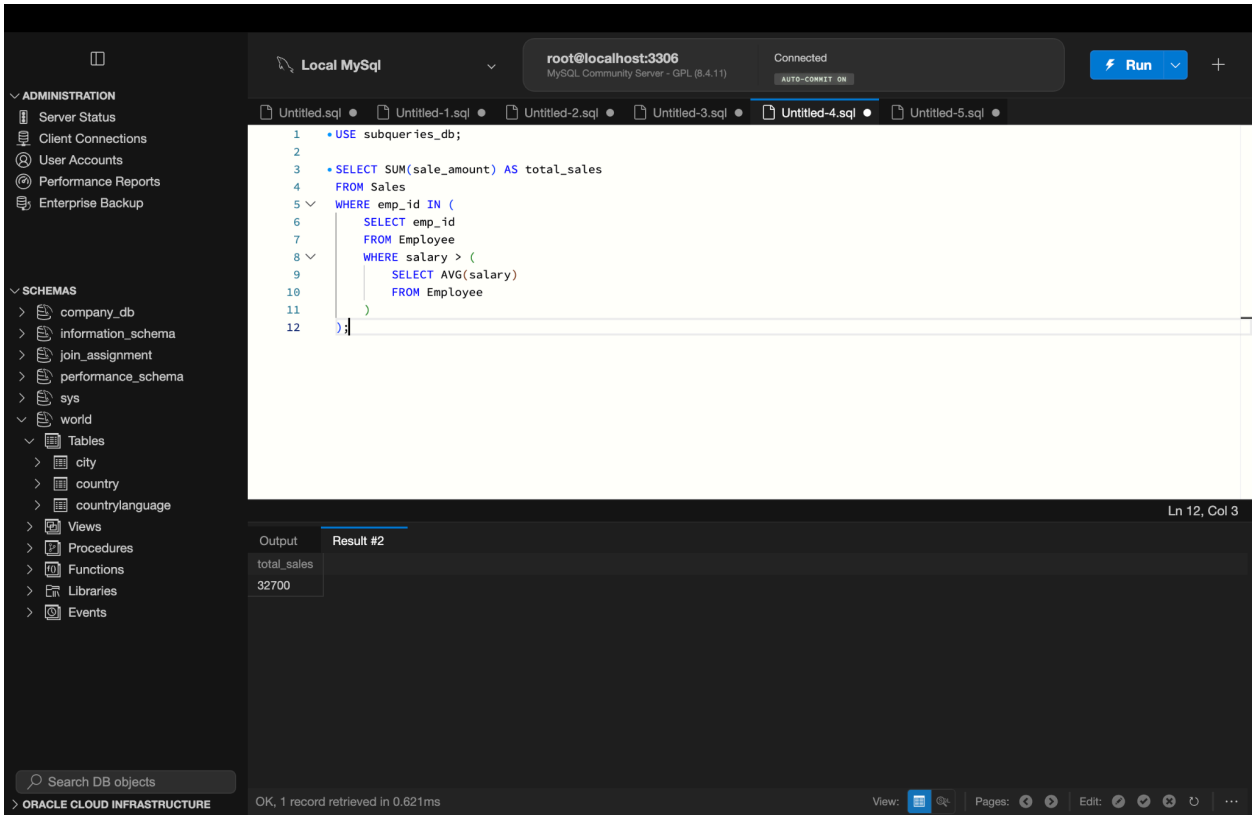
);
```

Explanation:

The subquery retrieves the employee IDs of employees who have made sales. The main query displays employees whose IDs do not appear in the Sales table.

Result: All 10 employees have made at least one sale, so no records are returned.

Practical Proof :



Question 12: List departments where the average salary is above ₹55,000.

SQL Query:

USE subqueries_db;

SELECT department_id, department_name

FROM Department

WHERE department_id IN (

SELECT department_id

FROM Employee

GROUP BY department_id

HAVING AVG(salary) > 55000

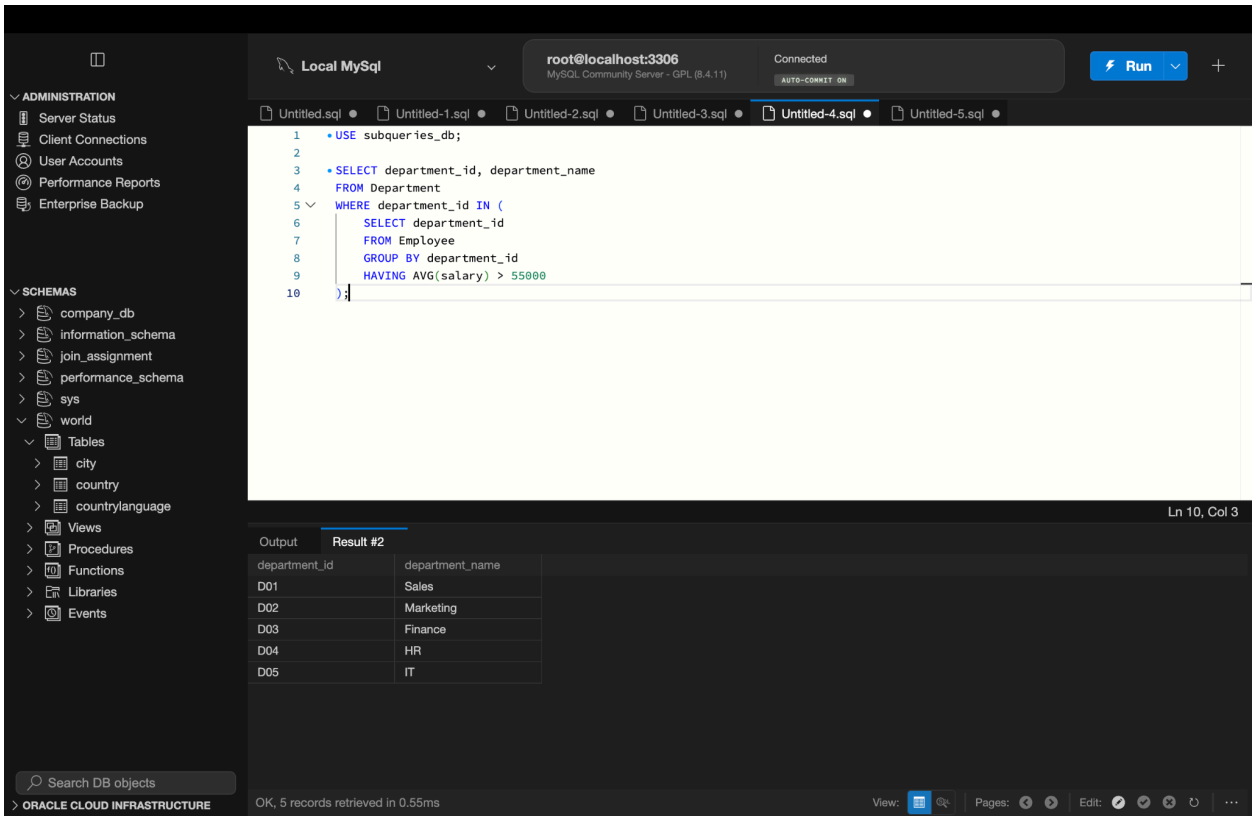
);

Explanation:

The subquery groups employees by department and calculates the average salary for each department. The `HAVING` clause selects departments where the average salary is greater than ₹55,000. The main query displays the corresponding department IDs and names.

Result: All five departments have an average salary above ₹55,000.

Practical Proof:



Question 13: Retrieve department names where total sales exceed ₹10,000.

SQL Query:

```
USE subqueries_db;

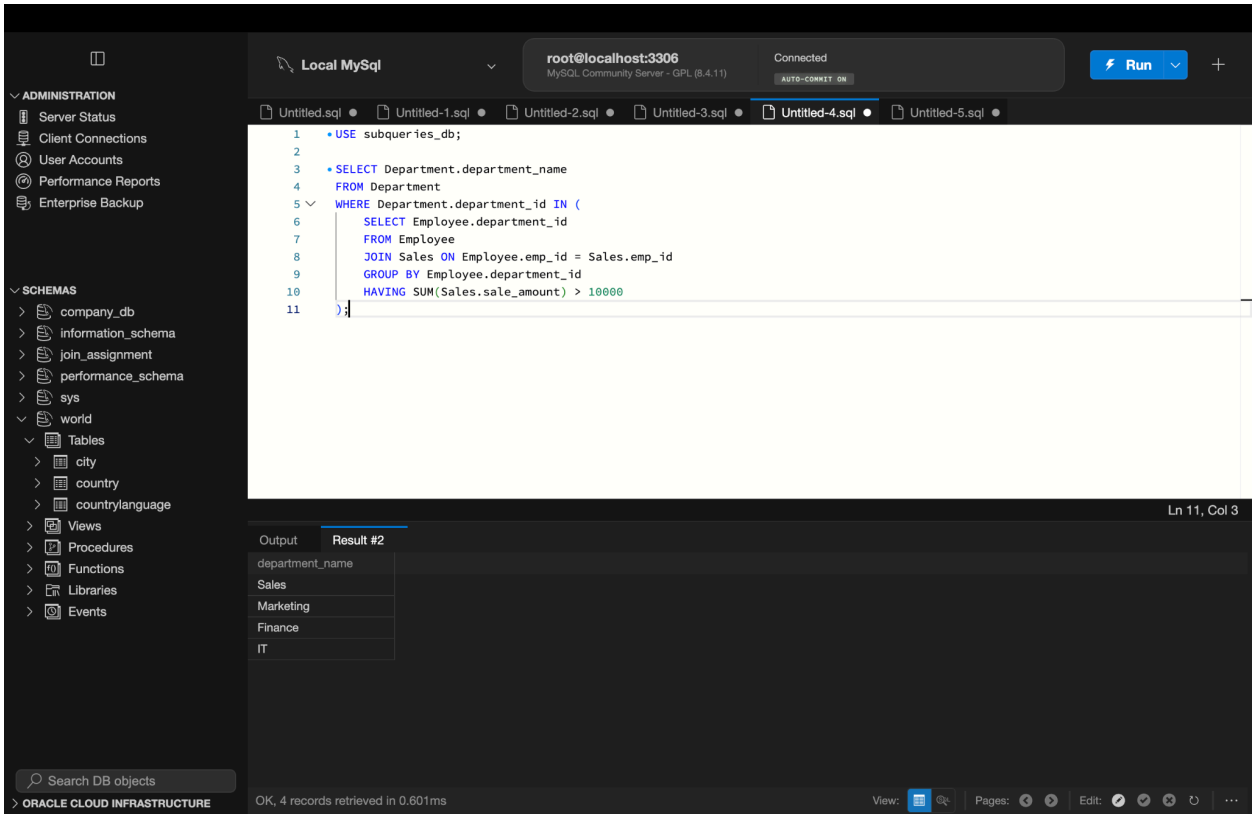

SELECT Department.department_name
FROM Department
WHERE Department.department_id IN (
    SELECT Employee.department_id
    FROM Employee
    JOIN Sales ON Employee.emp_id = Sales.emp_id
    GROUP BY Employee.department_id
    HAVING SUM(Sales.sale_amount) > 10000
);
```

Explanation:

The subquery joins the Employee and Sales tables, groups sales by department, and calculates the total sales for each department. The `HAVING` clause selects departments whose total sales exceed ₹10,000. The main query displays their department names.

Result: Four departments have total sales exceeding ₹10,000.

Practical Proof:



Question 14: Find the employee who made the second-highest sale.

SQL Query:

USE subqueries_db;

SELECT Employee.name, Sales.sale_amount

FROM Employee

JOIN Sales ON Employee.emp_id = Sales.emp_id

WHERE Sales.sale_amount = (

SELECT MAX(sale_amount)

FROM Sales

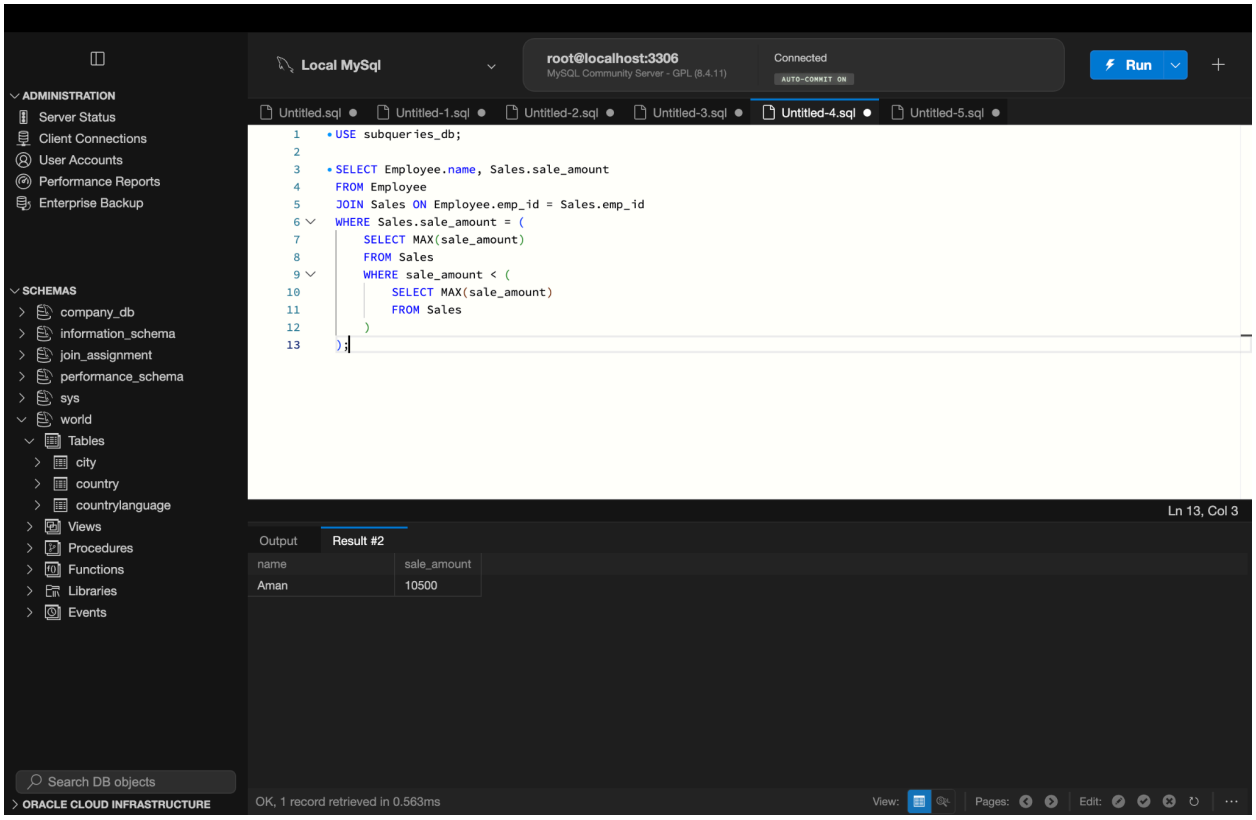
```
    WHERE sale_amount < (
        SELECT MAX(sale_amount)
        FROM Sales
    )
);
```

Explanation:

The innermost subquery finds the highest sale amount, which is ₹12,000. The next subquery finds the highest sale amount below ₹12,000, which is ₹10,500. The main query retrieves the name of the employee who made that sale.

Result: Aman made the second-highest sale of ₹10,500.

Practical Proof:



Question 15: Retrieve the names of employees whose salary is greater than the highest sale amount.

SQL Query:

```
USE subqueries_db;


SELECT name, salary

FROM Employee

WHERE salary > (

   SELECT MAX(sale_amount)

   FROM Sales

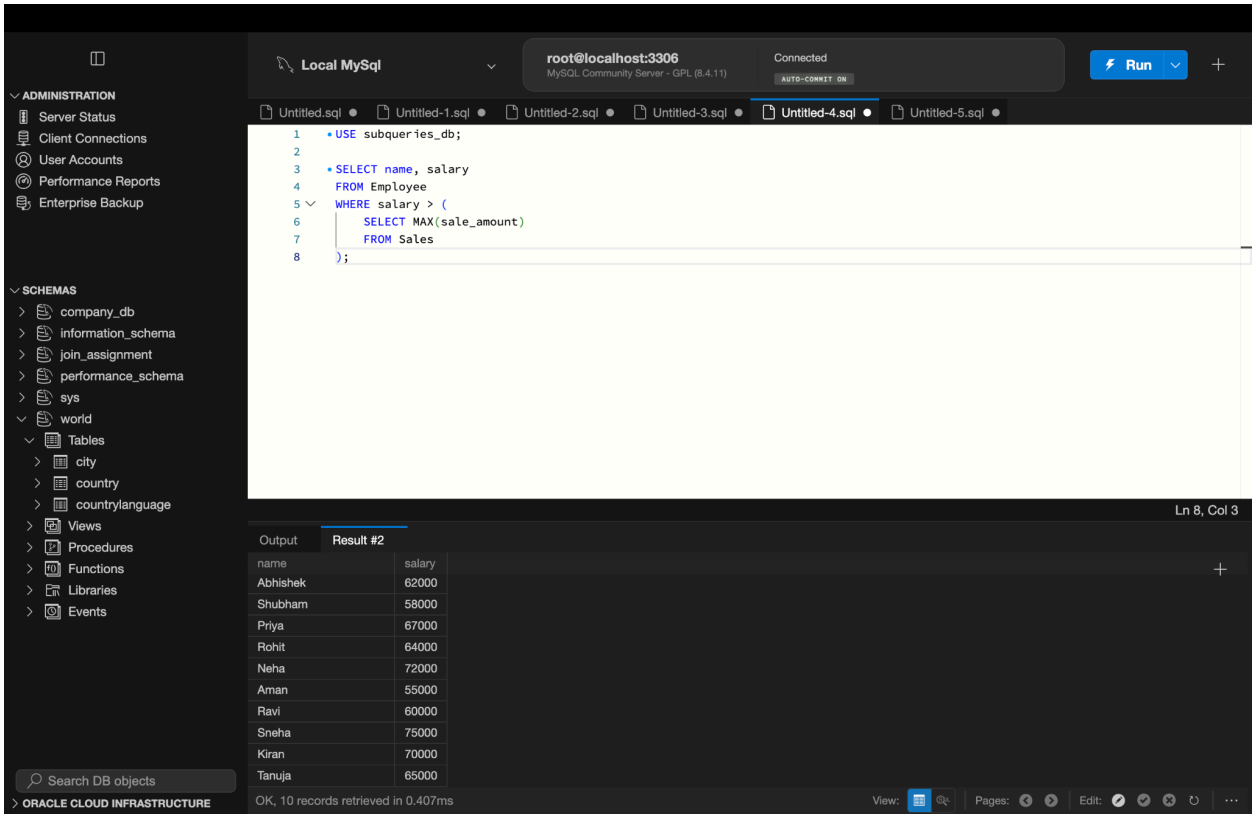
);
```

Explanation:

The subquery calculates the highest sale amount, which is ₹12,000. The main query retrieves the names and salaries of employees whose salaries exceed ₹12,000.

Result: All 10 employees have salaries greater than the highest sale amount.

Practical Proof:


Local MySql
root@localhost:3306
MySQL Community Server - GPL (8.4.11)
Connected
AUTO-COMMIT ON
Run
ADMINISTRATION
Server Status
Client Connections
User Accounts
Performance Reports
Enterprise Backup
SCHEMAS
company_db
information_schema
join_assignment
performance_schema
sys
world
Tables
city
country
countrylanguage
Views
Procedures
Functions
Libraries
Events
Search DB objects
ORACLE CLOUD INFRASTRUCTURE
Untitled.sql
Untitled-1.sql
Untitled-2.sql
Untitled-3.sql
Untitled-4.sql
Untitled-5.sql
1 USE subqueries_db;
2
3 SELECT name, salary
4 FROM Employee
5 WHERE salary > (
6 SELECT MAX(sale_amount)
7 FROM Sales
8);
Ln 8, Col 3
Output
Result #2
name salary
Abhishek 62000
Shubham 58000
Priya 67000
Rohit 64000
Neha 72000
Aman 55000
Ravi 60000
Sneha 75000
Kiran 70000
Tanuja 65000
OK, 10 records retrieved in 0.407ms
View:
Pages:
Edit: